

BERN BELPMOOS AP, Switzerland

WMO: 066300

Lat: 46.906N

Lon: 7.505E

Elev: 510

StdP: 95.35

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.2	-7.8	-14.0	1.2	-10.2	-9.2	1.8	-6.6	7.9	8.3	6.8	7.4	2.0	50	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	31.8	21.0	29.9	20.2	27.8	19.2	21.6	29.3	20.6	27.8	19.8	26.2	2.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.8	24.6	18.1	13.9	23.0	17.2	13.1	22.0	65.2	29.6	61.8	28.0	58.8	26.5	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 7.0	5.8	4.9	DB	-14.0	34.4	3.5	1.6	-16.5	35.6	-18.6	36.5	-20.6	37.4	-23.2	38.6	(4)
(5)			WB	-14.3	23.5	3.6	0.8	-16.8	24.1	-18.9	24.6	-20.9	25.1	-23.5	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	1.2	1.3	5.7	9.7	13.2	17.9	19.5	19.3	15.1	10.7	5.4	1.4		
	DBStd	7.57	3.89	4.28	3.26	3.51	3.49	3.30	3.11	3.12	3.11	3.44	3.37	3.80		
	HDD10.0	1158	274	245	136	47	11	0	0	0	1	36	143	266		
	HDD18.3	3175	532	478	392	258	163	48	22	27	104	238	389	524		
	CDD10.0	1184	1	0	3	39	109	236	294	288	154	56	4	0		
	CDD18.3	160	0	0	0	0	3	34	59	57	8	0	0	0		
	CDH23.3	1923	0	0	0	12	70	448	691	607	93	1	0	0		
	CDH26.7	630	0	0	0	1	8	152	250	209	10	0	0	0		
(14) Wind	WSAvg	1.8	1.9	1.8	1.9	2.0	2.1	2.0	2.0	1.8	1.6	1.6	1.6	1.8		
(15) Precipitation	PrecAvg	1277	69	73	81	98	141	146	148	143	108	92	90	87		
	PrecMax	1601	162	175	250	188	230	258	337	249	171	160	196	175		
	PrecMin	995	20	24	37	26	48	63	46	26	56	12	6	7		
	PrecStd	160	38	40	43	43	55	46	59	53	37	35	45	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	15.2	19.9	25.1	27.9	33.1	34.2	33.2	28.0	23.0	17.2	12.2		
		MCWB	10.1	8.9	11.4	15.1	17.4	22.5	21.8	20.9	18.8	15.2	12.5	8.9		
	2%	DB	10.2	12.8	17.9	22.2	25.1	30.9	31.9	31.0	26.1	20.9	15.0	10.9		
		MCWB	7.3	7.8	10.3	13.5	16.1	20.7	20.5	20.8	18.4	15.0	11.5	7.7		
	5%	DB	8.2	9.9	15.9	20.9	23.2	28.1	29.8	29.1	24.0	18.2	12.2	8.9		
		MCWB	6.1	6.3	9.4	12.7	15.7	19.6	19.8	20.1	17.3	14.1	9.6	6.6		
	10%	DB	6.2	7.1	13.1	18.1	21.1	25.9	27.1	27.0	21.9	16.2	10.2	6.9		
		MCWB	4.8	4.8	8.2	11.6	14.5	18.5	18.8	19.4	16.4	13.2	8.2	5.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	9.9	12.1	15.4	18.7	23.4	22.5	22.4	19.7	16.7	13.0	9.6		
		MCDB	13.1	14.1	18.1	23.4	25.4	32.2	31.5	29.8	26.0	20.7	16.8	11.9		
	2%	WB	8.2	7.8	10.5	14.3	17.5	21.4	21.2	21.5	18.8	15.5	11.6	8.2		
		MCDB	10.4	11.4	16.6	21.8	23.3	28.9	29.3	29.0	24.6	18.9	14.6	10.4		
	5%	WB	6.2	6.3	9.5	13.0	16.2	20.2	20.3	20.5	17.9	14.6	10.1	6.8		
		MCDB	7.8	9.2	14.8	19.5	21.9	26.8	27.4	27.6	22.8	17.8	12.3	8.6		
	10%	WB	4.9	4.9	8.6	12.0	15.0	18.9	19.5	19.8	16.9	13.6	8.8	5.2		
		MCDB	6.0	6.9	13.2	17.5	20.2	24.6	25.8	25.8	20.8	16.3	9.8	6.4		
(35) Mean Daily Temperature Range	MDBR		7.1	9.0	11.9	12.7	11.8	12.3	12.7	12.9	11.8	10.3	7.7	7.5		
	5% DB	MCDBR	9.5	14.3	17.8	18.4	16.4	16.6	17.0	16.5	14.9	13.8	11.7	9.8		
		MCWBR	7.0	9.5	10.7	9.8	8.3	7.7	6.8	7.1	7.6	8.7	8.1	7.2		
	5% WB	MCDBR	9.0	13.3	15.8	16.5	14.3	15.7	15.3	15.3	13.7	12.2	11.3	9.2		
		MCWBR	6.8	9.0	9.7	9.1	7.5	7.5	6.8	7.0	7.3	7.9	7.9	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.317	0.358	0.398	0.412	0.414	0.414	0.403	0.379	0.353	0.317	0.288		
	taud		2.415	2.342	2.283	2.216	2.230	2.272	2.268	2.309	2.358	2.412	2.449	2.430		
	Ebn at Noon		805	848	859	854	854	853	847	841	828	796	763	762		
	Edh at Noon		69	91	112	132	135	130	129	119	104	85	67	61		
(44) All-Sky Solar Radiation	RadAvg		1.26	2.11	3.28	4.36	5.01	5.93	5.77	5.02	3.89	2.49	1.37	1.02		
	RadStd		0.15	0.32	0.48	0.63	0.46	0.55	0.50	0.46	0.35	0.25	0.18	0.17		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (114 neighbors)		N/A	N/A	N/A	+0.75	+0.41	+0.31	N/A	N/A	+57	+16			

Nomenclature: See separate page